

SnowBat

Technical Report

ECE Senior Capstone Design

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I. Introduction

The purpose of this report is to document the technical design, implementation, and validation of SnowBat, an autonomous aerial search-and-rescue drone developed to accelerate avalanche victim localization in backcountry environments.

Context

SnowBat was inspired by two members of the team being backcountry athletes—one a skier, one a mountaineer—who recognized that avalanche rescue technology has not meaningfully changed in decades. Buried victims' companions are still required to conduct a foot-based search using a handheld 457 kHz transceiver, with survival probability dropping from 90% to 30% in 15 minutes, and organized SAR too far away to help. SnowBat is a purpose-built, companion-deployable autonomous drone that automates this search: the operator presses a single button, and the drone sweeps the debris field, tracks the victim's beacon signal, and localizes the burial site without any pilot input. The current prototype flies under radio control, the onboard beacon has been successfully decoded in real time, and the navigation algorithm has been written, though end-to-end autonomous beacon-tracking has not yet been validated in flight. This report documents the full technical design of the system and the validation work completed to date.

Scope

This report covers the full technical design of the SnowBat system. We first establish the baseline for why this product is needed and how this product will contribute value, then details the electrical, mechanical, and software subsystems of SnowBat. Finally, we go into detail on the process, including testing, timeline, and lessons learned. The report concludes with a summary of work completed and appendices.

II. Design Project Background

Background

Avalanches are a universally feared natural disaster due to their spontaneous and severe nature. A snow avalanche is defined as the rapid flow of snow down a slope [1]. The two most common types of avalanches are Slab and Loose Snow Avalanches, with Slab Avalanches posing the most severe threat to human life, accounting for 90% of avalanche fatalities [2]. Slab avalanches typically occur on 30-50 degree slopes and result from the accumulation of snow layers of differing densities over multiple storm cycles. Variations in snowpack cohesion, combined with environmental loading factors such as wind and rain, create internal weak layers. When subjected to external triggers--often human activity during winter sports--these weak layers collapse catastrophically. This causes the top snow slab to fracture and accelerate downhill, depositing metric tons of snow over a large search area and burying victims under meters of snow.

Due to the sheer amount of snow that the victim becomes buried in, an examination of the epidemiology of avalanche victims reveals asphyxiation and trauma are the two leading killers in an avalanche scenario. Asphyxiation is the leading cause of death, responsible for approximately 82% of fatalities, while trauma contributes to roughly 29% (with overlap between categories) [3]. Demographic analysis shows that 58.2% of avalanche deaths occur among backcountry and out-of-bounds skiers and climbers, whereas only 5.2% occur in-bounds, highlighting the positive relationship between mortality and the lack of rapid access to professional rescue resources [4]. With asphyxiation being our primary killer, the survival timeline is critical. Within 15, buried victims have a survival probability of roughly 90%, but survival drops to around 30% by 35 minutes, and approaches 0% beyond 90 minutes [5]. We define the first 15 minutes, where survival is above 90%, as the critical time window.

Various avalanche mitigation techniques are deployed in both front country (in-bounds, ski-resort) and back country (uncontrolled wilderness) environments. Front country locations are heavily trafficked and have ski patrols who monitor snow conditions multiple times per day. Avalanche cases in the front country rarely end in fatality. The nature of the front country allows for rescue to be achieved in the critical time window. Backcountry environments, on the other hand, are not heavily trafficked, unsurveyed, and hours away from organized rescue. A survey from Vail Resorts estimates that 1 in 4 backcountry skiers have been caught in an avalanche [6]. As a result, backcountry wintersports athletes have developed multiple standardized protocols. Essential parts of kit carried by all backcountry party members are a 457 kHz transceiver, snow shovel, and snow probe. A 457 kHz transceiver, or avalanche beacon, allows for party members to search and locate each other after an avalanche. The wavelength is chosen for its low attenuation in snow, allowing for penetration to reach about 70m in packed snow [7]. The following procedure is standardized by the American Avalanche Association (A3), Canadian Avalanche Association (CAA), and International Committee for Alpine Rescue (ICAR) [8].

1. Before entering the backcountry, all party members switch their transceiver to “SEND” mode.
1. In the case a party member is fully buried, the remaining members switch their transceivers to “SEARCH” mode to pick up the buried victim’s transmitting signal.
2. Start at the victim’s last seen point and work in a search pattern (zig zag, canvas, etc)
3. Once a signal is picked up, follow the directional cue
4. Within 3 meters, directional cues are unreliable so
5. Probe for the victim’s body using a snow probe
6. Dig to the victim using snow shovel

If the victim is extricated in the first 15 minutes, critical time window, their survival chance is >90%. This procedure highlights two primary bottle necks: the search phase and the extrication phase. Only anecdotal evidence alludes to the time it takes to locate the victim, but a 2019 study found that average extrication time for a victim fully buried under 1m of snow took an average of 7 minutes [9]. This means that our search time is restricted to only 13 minutes to maximize survival chance. Additionally, backcountry search and extrication must be executed by on-site companions as organized rescue has almost zero chance of reaching the victim in time.

An examination of current backcountry rescue practices and epidemiology of avalanche fatalities makes the limitation clear: even with transceivers, probes, and training, the search-and-rescue workflow consumes most of the 15-minute critical window. Because organized SAR cannot arrive quickly enough to influence survival, the only viable path to improved outcomes is to shorten the search or extrication phases to a matter of minutes. This gap strongly motivates the development of companion-rescue

tools--lightweight, deployable, and immediately available--that can rapidly localize and extricate a buried victim without relying on slow, foot-based search patterns or manual digging.

Customer Need Analysis

This project aims to design a lightweight, packable, and deployable aerial platform with an advanced sensor suite to enhance a companion rescuer's ability to locate a buried victim. Due to the hard-to-traverse and dangerous environment a post-avalanche snowfield becomes, we chose the air as a safe method of traversal that reduces the physical risk to the rescuer. Our device will combine standardized search technologies like the 457 kHz transceivers with sensorized autonomous flight, intelligent search patterns and snow penetrating radar to dramatically reduce search time.

An examination of our intended users: backcountry skiers, snowboarders, climbers, and mountaineers reveal several critical needs that must be addressed in our final product:

- Lightweight and unobtrusive
- Intuitive user experience for fast deployment under stress
- Reliable in cold, extreme alpine environments
- Robust user interface for use with gloves
- Sufficient operating time for single or multiple victim scenarios

On the technical side, we have identified several objectives and constraints:

- Total system weight under 1 lbs
- High elevation flight where air density is lower
- Sensor suite of: 457 kHz receiver, downward Time-of-Flight (ToF) height estimation, IMU, camera
- At least 15 minute flight time, validated in freezing degree environments
- Foldable or collapsible design
- Autonomous search patterns

Design Functionality

SnowBat is a purpose-built, companion-deployable autonomous drone designed to automate the coarse search phase of avalanche beacon rescue. The system consists of a collapsible carbon fiber quadcopter with a custom 3D-printed housing, weighing approximately 1.5 lbs and folding to a 7.0"x10.0" footprint for integration into a standard backcountry ski kit. Onboard sensors include an IMU, barometer, and five time-of-flight rangefinders. These readings are fed into a custom flight controller that autonomously plans and executes a beacon-tracking search path by continuously steering toward a 457 kHz signal, compatible with any standard avalanche transceiver. Deployment requires no pilot skill: the operator unfolds the four arms, orients the drone toward the last-seen position, and presses a single launch button; the drone then autonomously sweeps the debris field, localizes the buried victim, and alerts rescuers via onboard LED. We target the search of the buried victim within five minutes to preserve the majority of the critical survival window for extraction.

Project Deliverables

The following deliverables represent the full output of the SnowBat project, spanning two parallel tracks. The technical deliverables provide a complete foundation from which further hardware and software development can be continued. The business deliverables provide a complete foundation from which fundraising and commercialization efforts can be immediately launched.

Business and Market Research

The business development track produced a substantial body of research supporting the commercial viability of SnowBat that culminated in a large document and submission to HopStart. This includes background research on the avalanche rescue market, analysis of competitor and adjacent products, identification of similar validated markets, and an assessment of future market opportunities including ski resort B2B channels and consumer retail. Pricing analysis covers cost of goods sold, target sell price, and gross margin modeling. Primary market research was conducted via structured interviews with three backcountry skiers to validate user needs and willingness to pay.

Additional commercial deliverables include a 3-minute promotional video, an 3-minute investor pitch, and an 8-minute formal pitch deck presentation.

Technical

Technical deliverables cover the full design and implementation of the SnowBat engineering prototype:

- Solidworks CAD files for all custom mechanical components, including the v1 design and engineering test model
- GitHub repository containing all firmware, including the ArduPilot fork, beacon decoding firmware, and navigation algorithm
- Custom PCB schematic and board layout files
- 20-minute technical presentation covering the full system design
- Technical poster summarizing the system for the JHU Design Day showcase
- This technical report
- Weekly design update slides maintained throughout the semester, documenting design decisions, test results, and iteration rationale

The physical prototype: the SnowBat engineering model as well as spare parts and electronics are also delivered as the primary hardware contribution of the project.

Prior Art and Patents

Thorough research into the domain resulted in numerous avalanche search technology patents, many stemming from European winter sports companies, specifically ski resorts safety teams. The research conducted shows a strong proof-of-concept of the viability of our proposed system while showing a clear gap between current technologies and our solving our problem. One common standard across developing safety technologies for the snow sports industry is faster locating technologies in the deep snow. Many

large resorts have been exploring the concept of using long distance and long flight time AUVs that can lower the searching cost and speed over helicopters [11]. In a BBC article Roland Georges who president of the high mountain guides office in the French resort of Courchevel states “All the guides in Courchevel have Recco reflectors in their ski gear. However, it takes time to raise an alert and get a helicopter out.” Following this, he talked about how he believes drones could become indispensable for mountain guides heading off the beaten track, saying “Having seen how small and quick to activate drones can be, I would not be surprised if all guides are soon carrying one” [11]. This shows how even major corporations in the snow sports market are aware of the potential for an innovative backpackable drone solution.

Exclusive of Patent Information

As part of our prior art, we researched papers on avalanche burial incidents and the potential for rapid rescue technologies in order to better understand the constraints that arise in the harsh environment. Among these sources was the article indexed under PMID 31264903, which analyzes survival outcomes in avalanche victims and emphasizes how critical early detection and extrication are [9]. Its evidence that even modest delays drastically reduce survival probability reinforces the need for faster locating capabilities. We also examined the clinical practice guidelines indexed under PMID 37945433, which present expert consensus on best practices in avalanche rescue, including the prioritization of rapid searching and the reduction of overall response time [10]. These guidelines highlight ongoing challenges in quickly locating buried individuals and encourage the adoption of technologies that reduce the time to initial detection.

A third important source we found that helps quantify the risk was the paper “Avalanche Survival Rates in Switzerland, 1981-2020” by Rauch, Brugger, Falk, et al. [16]. This study analyzed 1,643 critically buried avalanche victims and found that the probability of survival declines sharply within the first 10 to 30 minutes of burial. While median rescue times have improved from 45 minutes in the early 1980s to 25 minutes in more recent years, many victims still do not receive assistance within the critical survival window. The findings demonstrate that immediate location remains a major challenge in backcountry settings.

Collectively, these sources confirm that fast location is vital, that existing rescue operations are often delayed, and that the professionals in the area and researchers recognize the need for faster search technologies. The evidence from both empirical studies and expert guidelines supports the viability of a compact, personally carried drone as a solution for backcountry parties.

Patents Search and Findings

WO2019011868 [12]

Class A63: Sports; games; amusements

Subclass A63B: Apparatus for physical training, gymnastics, swimming, climbing, or fencing; ball games; training equipment

Subgroup A63B29/02: Mountain guy-ropes or accessories, e.g. avalanche ropes; Means for indicating the location of accidentally buried, e.g. snow-buried, persons

Description: A large quadcopter that can be stored and deployed from the back of a vehicle such as a snowmobile which is used to search and locate buried persons in an avalanche that have an active transponder on them.

US11819737 [13]

Class A63: Sports; games; amusements

Subclass A63B: Apparatus for physical training, gymnastics, swimming, climbing, or fencing; ball games; training equipment

Subgroup A63B29/02: Mountain guy-ropes or accessories, e.g. avalanche ropes; Means for indicating the location of accidentally buried, e.g. snow-buried, persons

Description: Drone systems can include one or more drones that can fly. A drone can include a camera system that has one or more cameras. Drones can fly over an avalanche zone where an avalanche recently occurred to search for avalanche victims and mark the locations of avalanche victims.

EP3476734 [14]

Class B64: Aircraft; aviation; cosmonautics

Subclass B64D: Equipment for fitting in or to aircraft; flight suits; parachutes; arrangement or mounting of power plants or propulsion transmissions in aircraft

Subgroup B64D1/16: Dropping or releasing powdered, liquid, or gaseous matter

Description: The present invention relates to a drone for searching and marking a target comprising an aerial vector and also includes a dispensing device with several compartments containing a marking element and configured to release said marking element contained in a compartment when a target is detected so as to mark the location of said detected target.

III. System Design

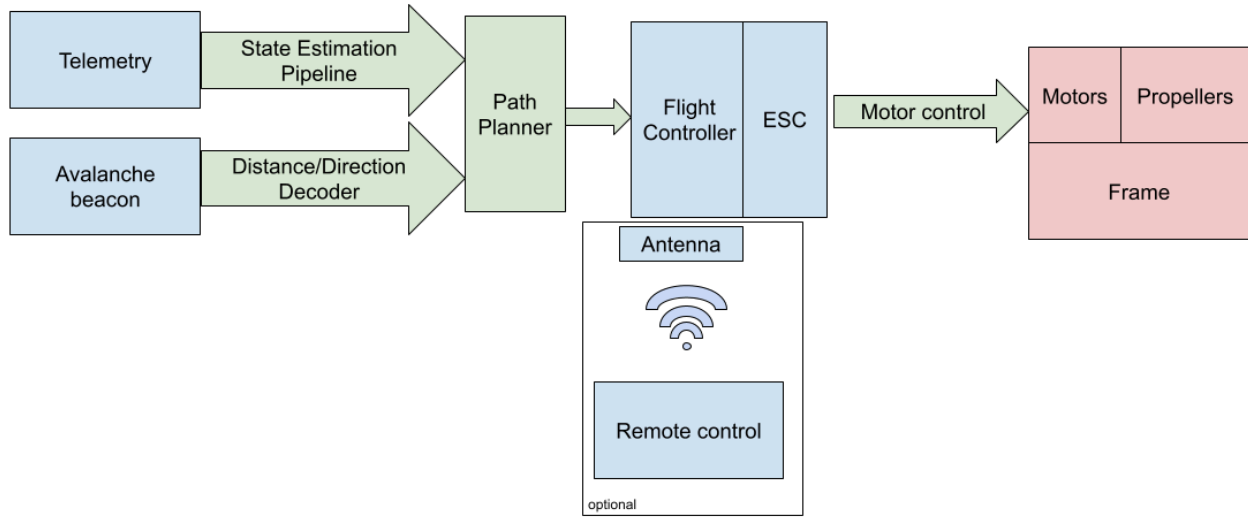


Fig. 1: Simplified block diagram of the overall system

The above diagram shows a simplified version of the major components of SnowBat relevant to its core purpose of autonomous search and flight. In blue are the major electrical components. In green are the major software components, and in red are the major mechanical components

At a high level, the goal of the drone is to fly to a series of waypoints that are either closer to the victim or will help collect information such that subsequent waypoints are more accurate. To achieve this goal, we have implemented a path planning algorithm that calculates waypoints based on the telemetry information and an avalanche beacon that is attached to the drone.

Telemetry includes distance measurements from four time-of-flight sensors, each angled downwards at a 45-degree angle, positioned around the base of the drone. The time-of-flight sensors are used to ensure the drone stays about 3 feet above the ground directly beneath it. Four of these sensors are needed to be able to do this on a sloped surface. The GPS gives us an absolute position and allows us to more accurately estimate waypoints. We put these readings through a state estimation pipeline to calculate the current position and orientation of the drone.

To avoid additional complexity, we obtain the distance and direction to the nearest avalanche beacon in send mode by reading straight from the LED display, which includes five arrows and two seven-segment digits.

We use the telemetry and displacement to the buried victim to inform our path planning algorithm, which runs on our flight controller. Once the next destination is calculated from our path planning algorithm, our

flight controller talks to the ESC, which ultimately controls the motors and flies the drone to the next waypoint.

Our propulsion system consists of four brushless motors mounted to the carbon fiber frame arms, each driving a two-blade propeller. The frame, motors, and 3D-printed housing together form the mechanical backbone of the system, designed to be lightweight, collapsible, and robust to the repeated crashes inherent to autonomous flight development.

Also included in the diagram is the optional remote control feature. This was an important feature for testing the prototype, but would not be included in the final product.

Mechanical Subsystem

The mechanical design of SnowBat was driven by five primary requirements: a total system weight under 1.5 lbs, a collapsed footprint small enough to fit in a standard backcountry pack, sufficient endurance for at least 15 minutes of flight, reliable operation at altitudes up to 10,000 ft and wind speeds up to 35 MPH, and a form factor that could physically accommodate the full sensor and electronics suite required for autonomous operation. These constraints were established up front and held as hard limits throughout the design process.

An additional practical constraint shaped the design significantly: the system needed to survive repeated crashes. Early-stage flight testing involves frequent uncontrolled landings, flips, and hard impacts. The mechanical design explicitly accounted for this, with an emphasis on impact-tolerant materials, replaceable components, and a structure that could be reassembled quickly after a failure so testing could continue the same day.

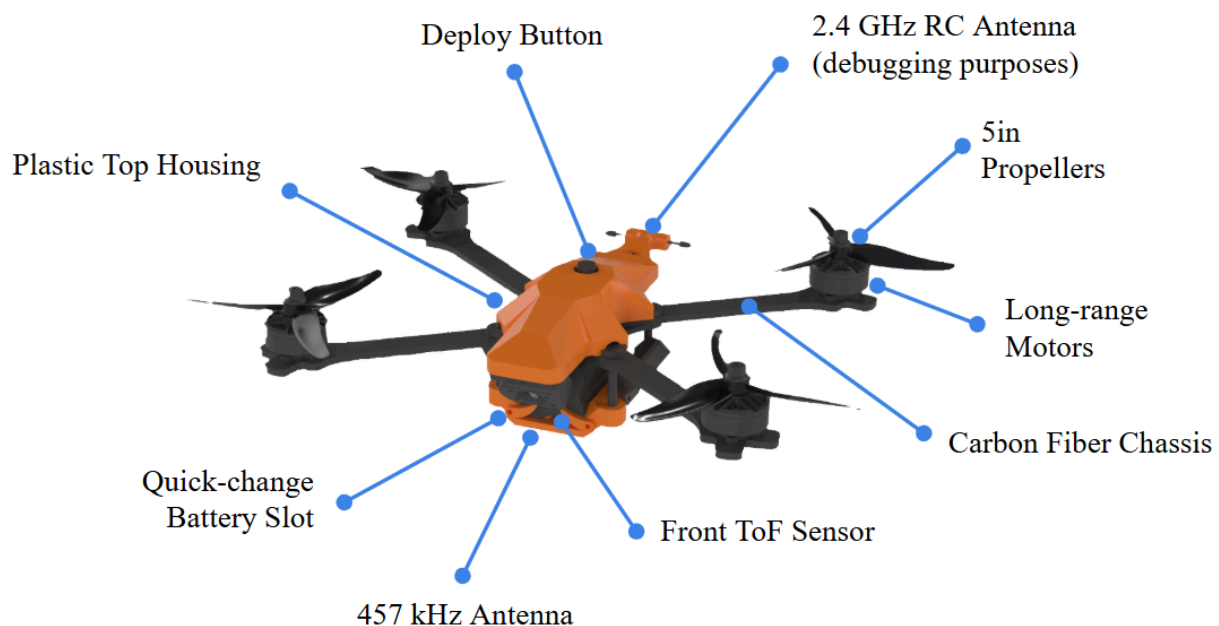


Fig 2: SnowBat mechanical design and parts integration.

Frame and Propulsion Selection

The primary structure is a Source One 7" carbon fiber long-range frame. Two propulsion configurations were evaluated: a 5" high-speed setup and a 7" long-range setup. Testing showed the 7" configuration provided substantially better hover efficiency at the target all-up weight, and the design was committed to that platform. Four 2808 1350KV brushless motors drive 7x4 two-blade propellers. The deployed footprint is 11.5"x14.5". The target stowed footprint is 7.0"x10.0" via collapsible arms, though this was not implemented on the engineering test unit.

All custom structural components were designed in Solidworks and manufactured via FDM printing in two materials. TPU was selected for parts expected to contact the ground and absorb impact: landing legs, undercarriage elements, and anything likely to take a hit during a crash or flip. Its flexibility allows these parts to deform under impact and recover, keeping the drone flyable across many test cycles without requiring parts to be swapped out. PETG was used for rigid structural parts and the housing shell. This two-material approach let the design be robust where it needed to be and stiff where it needed to be, without making the whole system out of a material optimized for only one of those conditions.

Integration

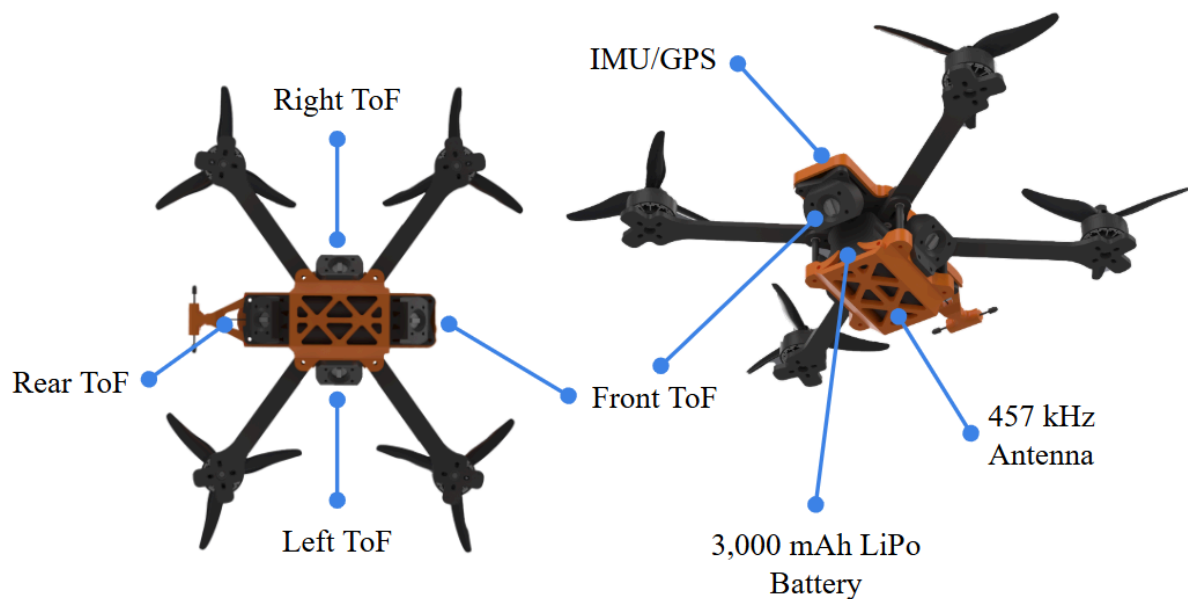


Fig 3: Integration of electrical subsystem.

The housing and undercarriage were designed to consolidate mounting for every subsystem in one place: five time-of-flight sensor positions (four populated), the custom PCB, flight controller, ESC, battery, 457 kHz antenna, 2.4 GHz RC antenna, and GPS module. Wiring routing for I2C, XSHUT, UART, and power

was factored into the housing geometry to give connectors accessible exit points and keep cables manageable during assembly.

Iteration

The design went through two full CAD revisions. The original SnowBat v1 prioritized a clean, enclosed form factor representative of the intended production configuration, shown on the left in Fig. 4. This version established the physical envelope and component layout.

Once full hardware integration began, the enclosed housing became a bottleneck. Reaching connectors, swapping components, and inspecting wiring required significant disassembly. The engineering test model opened up the enclosure with cutouts and access panels to support active testing and debugging, shown on the right in Fig. 4. Internal geometry was loosened so components could be pulled and reinstalled without taking apart the whole housing.

The v1 design remains the reference for the production form factor. All electrical and software testing was conducted on the engineering model.



Fig. 4: Evolution of SnowBat. Left: SnowBat v1. Right: SnowBat Engineering Test version.

Electrical Subsystem

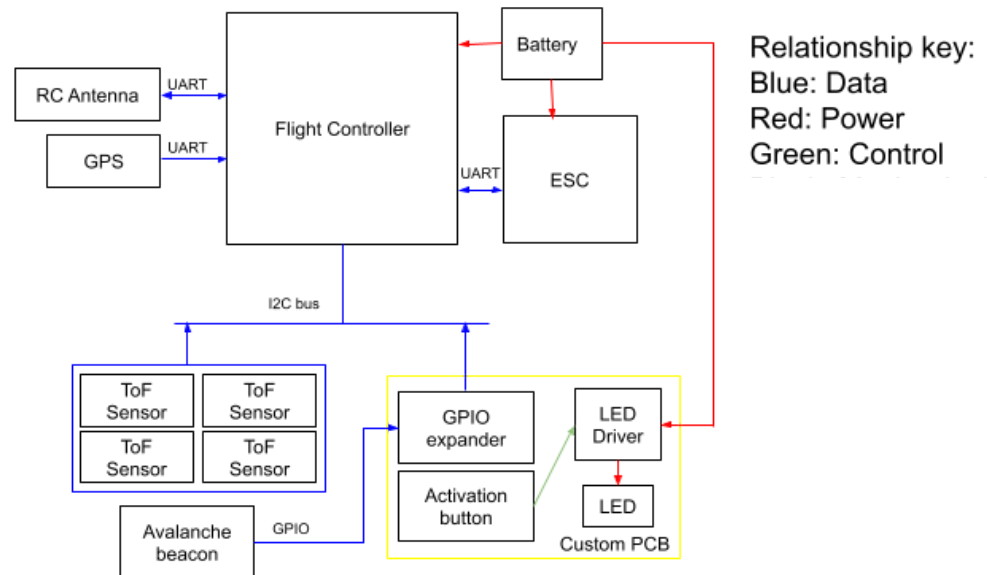


Fig. 5: Block diagram of the electrical subsystem

The electrical subsystem is responsible for power distribution, collecting telemetry, reading the avalanche beacon, and controlling the motors. All parts are directly mounted to the frame of the drone, except for the parts in the yellow box, which are first assembled onto a custom PCB and then mounted onto the drone.

Block Name	Function
Flight Controller	The brain of the drone. Run all the software, decide where to go, read telemetry and avalanche beacon. It also has a built-in IMU and gyroscope.
ESC	Control the motors based on what the flight controller decides
Time-of-Flight (ToF) Sensor	Measure the distance from the drone to the ground at various angles
Avalanche Beacon	Repurposed, stripped-down off-the-shelf avalanche beacon used to get the direction and distance of the buried avalanche victim.
GPS	Get (rough estimate of) absolute position of the drone
GPIO expander	Gives additional GPIO to the flight controller via I2C. Used to read the avalanche beacon's LED

	screen.
Activation button	A single, tall, IP67 button through-hole button used to activate the drone.
LED driver	A chip to provide constant current at 350 mA for the LED.
LED	A bright LED to help the user locate the drone after it has found the victim
Battery	A 4S 3000mA lithium ion battery that powers everything on the drone.
RC antenna	Used with the remote controller to allow manual control of the drone for testing purposes.
Remote controller (not pictured)	Used to manually fly the drone for testing purposes.

Table 1: List of all major electrical components.

Architecture Overview

The SnowBat electrical system can be separated into high-power propulsion electronics and low-power sensing and compute electronics. Some components are directly mounted onto the drone, and others were first put onto a custom PCB before being mounted to the drone. The directly-mounted components include the flight controller, ESC, battery, GPS, RC antenna, four ToF sensors, and the avalanche beacon. The custom PCB consolidates the GPIO expander, activation button, LED driver, and status LED into a single integrated board that interfaces with the flight controller over I2C.

At the center of the electrical system is the flight controller, which serves as the brain for the entire system. It runs all flight software and mission logic, performs sensor fusion using its built-in IMU and gyroscope, reads telemetry from the GPS and RC antenna over UART, communicates with the custom PCB over I2C, and issues motor commands to the ESC over a UART. The avalanche beacon connects to the custom PCB's GPIO expander via a 20-pin IDC connector, allowing the flight controller to read the beacon's LED display state and extract victim distance and direction information. Power flows from the battery directly to the ESC via an XT60 connector, and from the battery to the flight controller and all logic-level components via a regulated supply path.

Power Distribution

The entire drone is powered from a 4S, 3,000 mAh lithium-ion battery mounted to the frame. The 4S configuration provides a nominal voltage of 14.8V, which directly drives the ESC and motors. The flight controller and all logic-level peripherals are supplied from a regulated lower-voltage rail derived from the battery through the flight controller's onboard power circuitry. The 3,000 mAh capacity and motor selection are jointly sized to meet the 20-minute minimum endurance requirement at the target all-up

weight of 1.5 lbs, including margin for high-altitude operation where reduced air density requires higher motor throttle and consequently higher current draw.

Flight Controller and ESC

The flight controller is the brain of the SnowBat electrical system. It integrates a built-in IMU and gyroscope, which together provide the attitude and angular rate measurements necessary for stable flight. GPS position data arrives over a dedicated UART link, providing a coarse absolute position estimate used by the autonomous navigation algorithm. A second UART link connects the RC antenna, which interfaces with a remote controller used for manual flight and testing purposes. The flight controller communicates bidirectionally with the ESC over UART, sending per-motor throttle commands computed by the flight stabilization algorithm and receiving back ESC telemetry such as RPM and current draw. The ESC translates these commands into the three-phase PWM signals that drive each of the four brushless DC motors at the appropriate speed and direction. The motor control is borrowed from the open-source software Ardupilot, detailed in later sections. The flight controller also hosts the I2C bus master, over which it communicates with the custom PCB to read beacon data and ToF altitude measurements.

Custom PCB

The custom PCB (outlined in yellow in Fig 5) consolidates four functions onto a single board: reading the avalanche beacon's LED display via GPIO, managing the activation button input, driving the status LED, and providing the I2C interface back to the flight controller. The board connects to the flight controller via the shared I2C bus, to the avalanche beacon via a 20 GPIO connection, and to the battery power rail via a dedicated power input for the LED driver. The PCB is designed to be compact and mount onto and fit into the existing drone housing

The two GPIO expanders are used specifically to monitor the individual segment of the avalanche beacon's LED display, allowing the flight controller to extract the numeric distance and directional information at the beacon's native 1 Hz update rate.

The LED driver is a dedicated constant-current driver chip configured to supply 350 mA to the high-brightness status LED. Once the drone has localized the buried victim and landed, the LED activates to help rescuers visually locate the drone on the debris field. The activation button is a single tall-profile, IP67-rated through-hole button mounted externally on the PCB and accessible through the drone housing. The IP67 rating ensures reliable operation after exposure to snow, moisture, and freezing temperatures, and the tall-profile design allows positive actuation through standard ski gloves without requiring fine motor control.

Beacon Signal Decoding

The core sensing challenge of the SnowBat electrical design is extracting machine-readable distance and direction data from a commercial avalanche beacon. Rather than attempting to tap into the beacon's internal RF or digital processing pipeline, which would require proprietary knowledge of the beacon's firmware and/or low-frequency radio engineering, the approach taken is to tap directly into the beacon's

LED display driver lines, as shown on the left of Fig. 6.. The beacon used is a repurposed unit whose LED display outputs distance in meters and a discretized directional arrow at 1 Hz. After grabbing the state of each of the LED segments using the GPIO expanders, the flight controller then applies a software lookup table to translate the raw pin states into numeric distance and directional bearing values. This approach extracts the full information content of the beacon display without modifying the beacon's firmware or RF hardware, and requires no access to proprietary communication protocols.

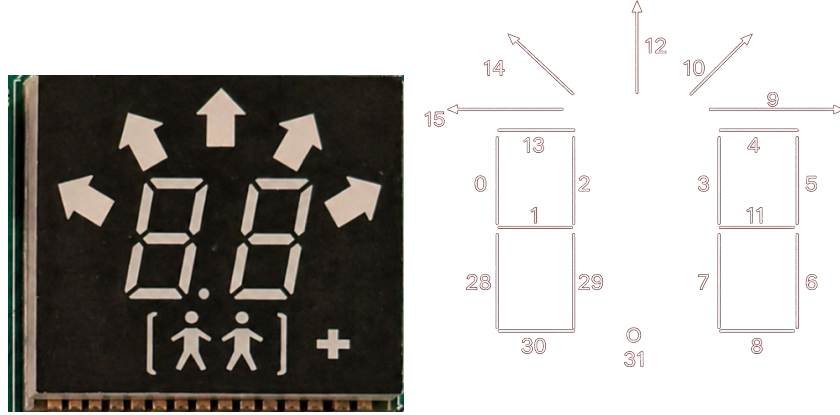


Fig. 6: The real beacon LED screen, and the final pin mapping used to decode the display.

Time-of-Flight Sensor Array

Altitude estimation and terrain-following are handled by an array of four VL53L0X time-of-flight distance sensors mounted directly to the frame and connected to the flight controller's I2C bus. Each VL53L0X sensor measures distance by emitting an infrared laser pulse and timing its return, with a maximum range of approximately 1,200 mm and low power consumption of 66 mW (20 mA at 3.3V), verified to function on snow surfaces. The sensors are mounted at 45° angles from the drone's horizontal plane (forward, backward, left, and right) so that each sensor's measured slant range can be projected onto the vertical axis using the following relationship:

$$h_i = d_i \cdot \sin \theta$$

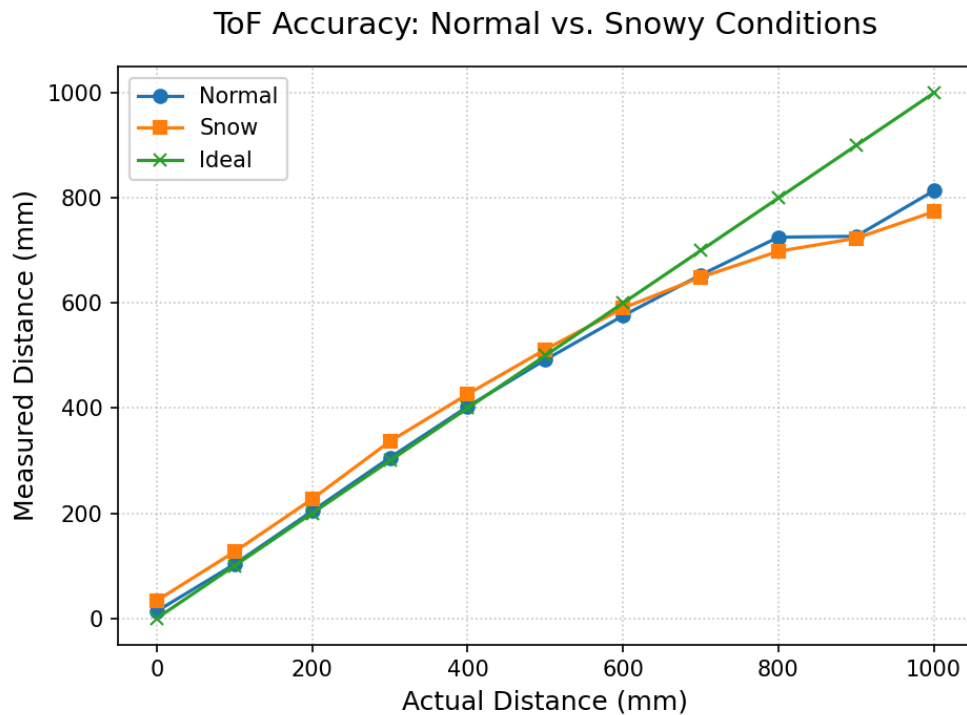
$$h = \sum d_i \cdot \sin \theta = \sin \theta \cdot \frac{d_1 + d_2 + d_3 + d_4}{4}$$

Fig, 7: ToF vertical height projection

The four projected height estimates are averaged to produce a single robust altitude measurement, mitigating the effect of any individual sensor being obscured or returning a spurious reading due to the irregular geometry of an avalanche debris surface. This arrangement yields a maximum hover height of approximately 2.78 feet (0.85 m) above the snow surface, which is sufficient for transceiver signal sampling while keeping the drone close enough to the debris field for accurate near-field localization.

Because all four VL53L0X sensors share the same default I2C address, sequential address reassignment is performed at boot time. All four XSHUT enable pins are first driven low via the GPIO expanders to disable all sensors simultaneously, then each sensor is powered on individually, assigned a unique I2C address, and held active before the next sensor is enabled. This results in four uniquely-addressed sensors on a single I2C bus.

Testing of the VL53L0X sensors on snow confirmed adequate performance within the operational altitude range. Below 800 mm, the sensor achieved an R^2 of 0.9823 on snow (compared to 0.9926 on normal surfaces), with an approximately 30 mm fixed offset on snow compared to 15 mm on hard surfaces. This offset is deterministic and compensated for in software. Performance degrades significantly beyond 800 mm due to reduced infrared reflectivity of snow at longer distances, which is the primary reason the operational hover altitude is constrained to remain within this validated range.



Fig, 8: Plot of the readings of the ToF sensor in normal vs snowy conditions.

Communication Buses and Data Flow

All digital communication is done via either I2C or UART. The I2C bus is controlled by the flight controller and connects to the custom PCB, over which the GPIO expander data and the four ToF sensors are read from. Separate UART interfaces connect the flight controller to three peripherals: the GPS module, the RC antenna, and the ESC. The GPS interface carries position fix messages, and is fused with the magnetometer and gyroscope via an extended Kalman filter (EKF) to estimate the drone's absolute yaw, which is fed to the navigation algorithm for coarse absolute positioning. The RC antenna interface carries pilot control inputs from a remote controller for manual testing, and would not be included in the final product. The ESC interface carries motor control and motor telemetry data.

Firmware/Software Subsystem

Our software system consisted of the firmware needed to interact directly with the sensors and bring up the custom PCB, as well as the software for waypoint navigation, which interprets the sensor data and decides which waypoints to fly to.

Firmware

The goal of the firmware is to provide basic functionality to the custom PCB and flight controller/ESC. It is an essential layer that bridges the hardware to the higher level software.

Beacon Decoding

This piece of firmware was written targeting our custom PCB. It is responsible for reading and interpreting the avalanche beacon's LED display state in real time. Two MCP23017 GPIO expander ICs, addressed at 0x24 and 0x20, are initialized over I2C with all 32 pins configured as inputs with internal pull-up resistors. The MCP23017s were implemented using the Adafruit MCP23017 library [17]. The main loop polls a dedicated trigger pin (MCP2 pin 15) that pulses in synchrony with the beacon's 1 Hz display update cycle. On each pulse, the firmware performs a bulk read of all 32 GPIO pins across both expanders, packing the results into a single 32-bit integer where each bit represents the active or inactive state of one display segment. Three segment subsets are then extracted from this integer using the `reformLED()` function, which reorders and concatenates the relevant bits according to pre-characterized pin-to-segment mappings: one subset for the left digit, one for the right digit, and one for the directional arrow indicator. The left and right digit bitmasks are each compared against a standard 7-segment lookup table to resolve the numeric distance value displayed on the beacon. The direction subset is read as a 5-bit encoded value corresponding to the discrete set of possible arrow directions shown on the beacon display. The decoded distance digits and direction value are then made available to the navigation algorithm, which uses them to compute the next waypoint toward the buried victim.

Ardupilot/Betaflight

In order to speed up the development process, we used common open-source softwares to provide the basic functionality for liftoff. After assembling the drone components, we immediately validated our setup with the default/pre-installed BetaFlight firmware [18] on our F745 Blitz Mini flight controller. By doing so, we were able to test our hardware at a high-level, specifically by connecting our RC antenna and radio controller through BetaFlight's interface, debugging some trivial yaw offset and initialization errors that would cause the drone to tumble/rotate along one of the arm axes, and tuning our PID constants appropriately for smooth flight. While the BetaFlight web/desktop interfaces were great for easily flashing different firmware versions and defining high-level behavior such as enabling our GPS module, radio controller buttons/modes, and testing our motors, we utilized a ground control station (GCS) called MissionPlanner to alter variable values and attempt connecting to our ToFs. Although we tested our ToFs individually, we ran into issues with defining their connection to our flight controller through MissionPlanner.

Most importantly, however, our hardware and beacon setup could not be integrated by default with the BetaFlight firmware. For more custom integration with our various devices over I2C, we switched to building our own firmware by forking Ardupilot [19], which contains a completely customizable C++ repository with support over countless vehicles. We specifically modified the ArduCopter codebase, perfect for our quadcopter setup, to integrate our custom hardware and search algorithm.

Time-of-Flight (ToF) Sensor

The first step of our code initialization involved distinguishing our ToFs with unique I2C (software) addresses on bootup. We used the Pololu library [20] to interact with the sensors for testing. Since each ToF came with a default I2C address of 0x29, we inserted code that sequentially interfaces with each XSHUT pin to “enable” sensors with different addresses (as shown below in Fig 9).

```

1 #include "Adafruit_VL53L0X.h"
2
3 const int SENSOR_COUNT = 4;
4 int xshutPins[SENSOR_COUNT] = {52, 50, 48, 46};
5 int sensorAddresses[SENSOR_COUNT] = {0x30, 0x31, 0x32, 0x33};
6
7 Adafruit_VL53L0X lox[SENSOR_COUNT]; // Only ONE object to rule them all
8
9 void setup() {
10     Serial.begin(115200);
11
12     // Put all sensors in reset
13     for (int i = 0; i < SENSOR_COUNT; i++) {
14         pinMode(xshutPins[i], OUTPUT);
15         digitalWrite(xshutPins[i], LOW);
16     }
17     delay(10);
18
19     for (int i = 0; i < SENSOR_COUNT; i++) {
20         digitalWrite(xshutPins[i], HIGH);
21         delay(10);
22
23         if (!lox[i].begin(sensorAddresses[i])) {
24             Serial.print(F("Failed to set address for sensor #"));
25             Serial.println(i);
26             while (1);
27         }
28         delay(10);
29     }
30
31     Serial.println(F("All sensors addressed. Starting measurements..."));
32 }
33
34 void loop() {
35     VL53L0X_RangingMeasurementData_t measure;
36
37     for (int i = 0; i < SENSOR_COUNT; i++) {
38         // Point the single object to the i-th sensor's address
39         // This doesn't re-initialize the hardware, it just updates the target I2C address.
40         lox[i].setAddress(sensorAddresses[i]);
41
42         lox[i].rangingTest(&measure, false);
43
44         Serial.print(F("S"));
45         Serial.print(i);
46         Serial.print(F(": "));
47         if (measure.RangeStatus != 4) {
48             Serial.print(measure.RangeMilliMeter);
49             Serial.print(F("mm  "));
50         } else {
51             Serial.print(F("---  "));
52         }
53     }
54     Serial.println();
55     delay(50);
56 }

```

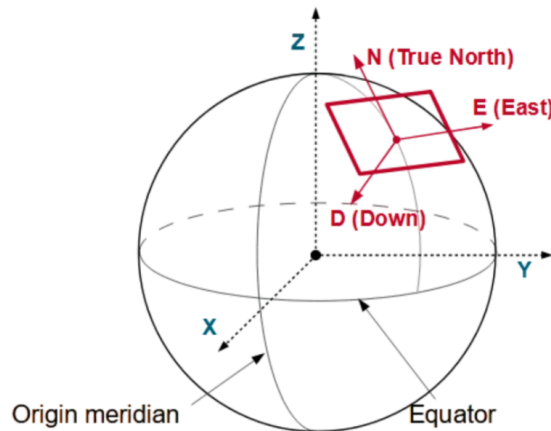
Fig. 9: ToF I2C address initialization

Software

The goal of the software was to provide waypoint navigation and coarse search functionality. It builds off of exposed state-estimation interfaces by integrating the required math for our proposed search algorithm(s).

Waypoint Navigation

For waypoint navigation, Ardupilot internally represents world coordinates as North-East-Down (NED), which is a common convention for many autonomous vehicles. Fig 9 shows a nice illustration of this coordinate frame:



Fig, 10: NED coordinate frame illustration - a localized “tangent plane” to the WGS84 ellipsoid

Importantly, aircraft navigation is often expressed in terms of heading over the Earth: north, east, and altitude [21]. The NED coordinate system makes horizontal motion intuitive, with the x and y coordinates representing ground-plane movement and z representing vertical movement in the downward direction.

Ardupilot’s built-in flight mode, “Guided”, comes with an easy-to-use interface for autonomous waypoint navigation that, among other things, accepts a destination NED coordinate. Our custom code utilizes this existing interface to build out our navigation functionality.

For our waypoint navigation search algorithm, we first needed to convert our target destination read from the beacon to an NED coordinate. As shown in lines 27 and 28 of Fig 11 below, we calculate the appropriate projection in the North and East directions from the received beacon distance/direction readings, and add that offset to the drone’s current NED coordinate. Since the Down (z) coordinate is entirely dependent on our ToF height readings, as described in detail above, and does not alter our ground-plane displacement locally, we set the Down component at the very end of the navigation pipeline and right before the method call to send the drone.

```

14 Vector3f beacon_reading_to_ned_coord(const Vector3f& current_ned,
15                                     float drone_yaw_rad,
16                                     float direction_rad,
17                                     float distance_m,
18                                     float max_step_m)
19 {
20     // Cap the step size. This is especially useful during the beginning
21     const float step_m = min(x: distance_m, y: max_step_m);
22
23     // Convert body-relative direction to world/NED direction
24     const float world_direction_rad = drone_yaw_rad + direction_rad;
25
26     Vector3f target;
27     target.x = current_ned.x + step_m * cosf(x: world_direction_rad); // North
28     target.y = current_ned.y + step_m * sinf(x: world_direction_rad); // East
29     target.z = current_ned.z; // Down unchanged here... modified later with ToF data
30
31     return target;
32 }

```

Fig, 11: NED coordinate calculation - Beacon target destination to NED coordinate for waypoint navigation.

Beacon Antenna Properties

As a side note, the beacon antenna is only a few centimeters long, meaning it is electrically tiny ($\ll \lambda$).

$$\lambda = \frac{c}{f} = \frac{3 \cdot 10^8}{4.57 \cdot 10^5} = 650m$$

Fig, 12: Emergency search signal wavelength

As Fig 12 shows above, the beacon emits an emergency search frequency at 457 kHz, which corresponds to a 650m wavelength. However, antennas behave optimally when their length is a specific fraction of the captured signal's wavelength: $\frac{1}{2} \lambda$ for standard dipoles. Since the beacon's antenna length is a few centimeters while the optimal length is 325m, the actual behavior is more similar to a magnetic dipole field, where flux lines form closed loops around the beacon.

Waypoint Navigation Continued...

Another important aspect to note is how we limit the step size with a `max_step_m`, in Fig 11, of our choosing. This is especially useful because, at the beginning of a typical search, large distance readings from the beacon can cause proportionally large waypoint jumps that would veer the drone off the flux line and further away from the buried victim. By setting the maximum step size to a relatively small value, we can repeat the waypoint generation at many discrete points along the flux lines and, therefore, keep the drone on track.

Of course, once we command the drone to a target NED coordinate, we must continuously check the Euclidean distance between the current and target NED vectors, as shown in Fig 13, to determine when to

generate the next waypoint. Once this L2 norm is within a defined tolerance, we proceed with calculating the next waypoint vector.

```
34 float compare_neds(Vector3f current_ned, Vector3f target_ned, bool use_down) {  
35     float dN = target_ned.x - current_ned.x;  
36     float dE = target_ned.y - current_ned.y;  
37     float dD = target_ned.z - current_ned.z;  
38  
39     if (use_down) return dN*dN + dE*dE + dD*dD;  
40     return dN*dN + dE*dE;  
41 }
```

Fig, 13: L2 error norm of source/target NED vectors

Search Algorithm

The first and most obvious approach for the search algorithm simply involves following the flux lines until the drone reaches the buried victim, as outlined in this section. This fast/easy approach follows how a human rescuer would attempt to find a victim, but is of course much faster and more reliable due to the drone's speed advantage and algorithmic nature.

An interesting extension of the search algorithm would involve a more machine learning adjacent approach. Instead of blindly following flux lines, which often lead the rescuer away from the victim initially only to loop back around towards them, we propose a gradient descent process where the drone moves in the direction of steepest decrease in distance to the victim. This could alternatively be framed as moving in the direction of steepest increase in signal intensity. Either way, the drone would need to move circularly to constantly monitor the true direction of the victim. This approach basically corresponds to the drone moving orthogonally to the flux lines, and would result in a much straighter and more direct path to the buried victim.

IV. User Interface Guide

The SnowBat user interface is intentionally minimal. The system is designed to be deployed under acute stress, in freezing temperatures, by a user wearing ski gloves who may have just witnessed a companion get buried. Complexity in the interface directly translates to seconds lost. The entire user-facing interface consists of two elements: a single button and a single LED.

The operator interaction at deployment is three steps: unfold the four drone arms, orient the drone toward the last known position of the buried victim, and press the launch button. From that point the system operates fully autonomously: no radio controller, no app, no configuration required. The launch button is a tall-profile, IP67-rated through-hole button mounted externally on the drone housing. IP67 ensures reliable operation after exposure to snow, moisture, and freezing temperatures, and the tall profile allows positive actuation through standard ski gloves without requiring fine motor control. During flight, the onboard LED pulses at a frequency proportional to the drone's proximity to the victim: a slow pulse at range, accelerating as the drone closes in, and steady on once the victim has been localized and the drone

has landed. This gives rescuers a continuous visual indication of search progress from the moment of launch.

The RC antenna present on the current engineering prototype is for testing use only and is not part of the intended user interface for the final product.



Fig 14: 3-step process for using SnowBat. 1. Orient SnowBat and press the “deploy” button. 2. SnowBat autonomously locates buried victim. 3. Rescuers dig victim out.

V. Risk Reduction

This project involved several major technical risks due to the combination of autonomous flight, custom electronics, avalanche beacon integration, and goal of operation in snowy outdoor environments. Throughout the semester, a large focus of the project was reducing these risks early through subsystem testing, prototyping, and parallel development before final integration into the drone prototype.

One of the project risks was determining whether the selected sensors and specifically the ToF sensors would function reliably in snow conditions. Since the final system depended on maintaining a low and consistent altitude over uneven avalanche terrain, the sensors needed to provide stable readings in snowy conditions. To reduce this risk, the ToF sensors were tested independently using ESP32s before integration with the flight controller. Testing was done on both normal surfaces and snow to compare accuracy, range, and consistency. These tests confirmed that the sensors performed reliably within the intended operating range and that the measurement offset caused by snow could be compensated for in software along with basic filtering.

Another technical risk involved decoding the avalanche beacon display. After our initial research phase, instead of attempting to directly interface with the beacon’s RF hardware or internal firmware, which would have added significant complexity and uncertainty, we decided to focus on decoding the LED display outputs. Before designing the custom PCB, the GPIO extender chips were tested independently using breakout boards and ESP32 development boards to ensure that the display segments could be read reliably over I2C. Once this communication was validated, additional testing was done to decode the beacon’s directional arrows and seven-segment distance displays into usable digital values. These experiments confirmed that the GPIO expanders were fast and reliable enough for the application and significantly reduced the risk of PCB redesign later in the semester.

Risk reduction also influenced how the project schedule was organized. To avoid waiting for the custom PCB to arrive before continuing development, flight testing was intentionally separated and performed

concurrently. While the PCB was being designed and manufactured, flight testing and flight-controller configuration work continued independently. During this period, the drone platform was tested thoroughly using BetaFlight to find stable PID values and evaluate its overall flight behavior. Two motor configurations were also tested during this phase: 5-inch higher-speed propellers and 7-inch more efficient ones. Since stretching flight time was one of the major project goals, testing showed that the 7-inch propellers provided sufficient stability and lift while significantly improving efficiency compared to the 5-inch design.

Lastly, early flight testing in BetaFlight allowed our team to better understand tuning requirements, frame behavior, and sensor placement before transitioning to ArduPilot for autonomous flight development. While hardware testing and PCB fabrication were ongoing, work also began on waypoint generation, telemetry handling, and autonomous flight logic in ArduPilot. This overlap in development reduced downtime and prevented software development from becoming blocked by hardware delays.

Overall, the project used continuous subsystem validation, parallel development, and iterative prototyping to reduce technical and schedule risk throughout the semester. By independently validating sensors, GPIO expanders, beacon decoding, propulsion configurations, and flight-controller behavior before final integration, the team was able to significantly reduce uncertainty in the final system design.

VI. Prototype Cost

The total prototype cost for SnowBat is \$1,630.24, sourced across specialty FPV retailers, general electronics distributors, general retailers, and PCB fabrication services. This figure reflects the full cost of the engineering prototype including all hardware iterations: multiple motor configurations were evaluated, ToF sensors were ordered across three separate purchases as the design evolved, and several components were ordered in quantity to account for expected attrition during crash testing. A detailed itemized BOM is provided in Appendix B. The table below summarizes costs by category.

The custom PCB was designed by the team and submitted to JLCPCB for fabrication. PCB fabrication and international shipping introduced the longest lead time in the project at approximately 2–3 weeks, which was accounted for in the project schedule by beginning PCB design early and placing the order as soon as the schematic was finalized.

The majority of propulsion and avionics components were sourced from specialty FPV retailers, which carry the specific motor, stack, and frame hardware used in the design. These vendors typically ship within 2–3 business days domestically, though one motor order (T-Motor Velox V3 2306, Pyrodrone) noted limited stock of only 2 units at time of purchase, requiring a second order from an alternative vendor. The avalanche beacon was sourced from SnowBigDeal, a specialty outdoor retailer. Sensing and PCB components were sourced from Adafruit and Digikey, both of which carry standard lead times of 1–3 business days for stocked parts. One unaccounted-for lead time early in the project was the ESC, which shipped directly from a factory in Hong Kong and took approximately two weeks to arrive. Additionally, purchases made through JHU's procurement process added a few days of internal processing time on top of vendor lead times. Both of these delays were identified early and factored into the ordering schedule for subsequent purchases.

Equipment and tooling costs were not a significant line item for this project. Electrical development and testing, including oscilloscope measurements, continuity checks, and bench power supply work, was conducted in the Wyman Building basement lab using equipment available through the ECE department. Hand tools used during mechanical assembly and hardware integration were sourced from teammates personally. FDM printing of all custom TPU and PETG components was done at the JHU Makerspace, which provides access to FDM printers at no per-print cost to students. No equipment purchases were required to complete the prototype.

VII. Test Plan

The testing process was designed around gradual subsystem validation followed by staged full-system integration. Since the project combined autonomous flight, terrain and obstacle sensing, avalanche beacon localization, and custom electronics, testing has to be done continuously. This testing process directly supported the risk reduction stated above in creating concurrent design tasks throughout the semester.

The first stage of testing focused on validating the drone platform itself before integrating custom electronics. Initial flight tests were conducted using BetaFlight to tune PID values, verify stable flight behavior, and evaluate the general handling characteristics of the drone. During this process, both 5-inch and 7-inch propeller configurations were tested to compare responsiveness, efficiency, and lift capability. Since long flight endurance was a required design goal, the 7-inch propeller was chosen after testing and thorough PID tuning showed it was possible to have sufficient stability with our design. This improved the drone's efficiency and expected flight time compared to the smaller propellers.

At the same time as the flight testing, the sensor systems were tested independently using ESP32 development boards. This allowed sensor debugging and firmware development to continue without interfering with flight-controller testing. The VL53L0X ToF sensors were tested on snow and normal surfaces to determine their usable operating range and overall measurement consistency. These tests verified that the sensors performed reliably within the expected hover height of the drone and that software compensation could account for small offsets caused by snow reflectivity. This also allowed us to test various filtering methods to determine what might be required for consistent and precise values in the various possible snow and ice conditions. Also, testing for the avalanche beacon interface was done before the custom PCB was manufactured. The MCP23017 GPIO extender chips were first tested individually over I2C to verify stable communication to an ESP32 device over its standard libraries and an initial program was able to be made to confirm that they could reliably read the beacon display's values. After this, the avalanche beacon display itself was connected to the GPIO extender breakout boards so that the directional arrows and seven-segment displays could be decoded. This testing allowed the pin mapping and lookup table logic to be finalized before PCB fabrication, saving time later and also reducing the likelihood of major hardware revisions, such as needing to redesign the PCB, later in the process.

Once the custom PCB arrived, electrical testing focused on validating communication and integration between all devices. This included continuity checks, power verification, I2C communication testing, GPIO validation, and confirmation that the beacon decoding still functioned correctly on the integrated

PCB. After minor adjustments, due to the thorough earlier testing this was able to seamlessly integrate into the overall system as the prototype boards had to previously. Testing was also performed on the ToF sensors to ensure all of the sensors could be on the shared I2C bus that the PCB used and still properly communicate with the flight controller. During this time it was also confirmed that all voltage rails were properly maintained to all of the components when the drone was powered on.

Software testing was performed incrementally alongside the hardware integration process. Early testing focused on telemetry configuration and calibration, along with connectivity of peripherals through the BetaFlight software. Once the initial BetaFlight testing phase was completed, development transitioned into ArduPilot so that autonomous waypoint behavior and sensor-driven navigation could be tested. During this phase, telemetry input handling, waypoint calculations, and autonomous flight logic were created and integrated into the existing autonomous flight software contained in ArduPilot.

Overall, the testing process emphasized continuous iteration and subsystem validation throughout the semester. By testing sensors, communication systems, propeller and frame configurations, and flight software independently before full integration, we were able to identify issues early and reduce the risk where possible of major failures during the final stages of the project.

VIII. Project Schedule

The project schedule was divided into two major phases: the prototyping phase and the final build phase. A focus with our overall schedule was to allow for parallel development so that multiple parts of the project could progress simultaneously throughout the semester instead of waiting for one subsystem to be completely finished before starting another.

At the start of the semester, the focus was on research and planning our whole system breakdown. Our goal was to do the mechanical research, electrical and RF research, and software architecture planning all at roughly the same time. We worked for the first few weeks to better understand avalanche drone components we would need to order and where integration would be needed between them. The drone constraints we based our design off of were battery requirements and motor requirements for our weight and goal flight time, sensor options for working in snow, and the challenges associated with 457 kHz avalanche beacon connectivity. After the initial research stage of the first two weeks and ordering components for testing, we moved into our prototype of each subsystem. We began our first set of goals which were creating a simple version of the drone to test flight capability while also testing our sensor choices and customer firmware for connectivity with our transceiver separately.

Initial prototype flights were a major part of this first prototype for the project. Once we had received all of our parts, we had created two separate configurations for testing, a low speed 7 inch propellor design and a high speed 5 inch propellor design. With the goal of efficiency, we first determined whether 7 inch propellers were viable before deciding if it would be necessary for the more responsive 5 inch design. Multiple iterations of sensor mounting and light weight drone frame designs were done during this time to eliminate the chance of it being a block to the flight testing. During this physical build process we also had begun configuring our flight controller in the BetaFlight software for initial flight testing and PID tuning.

At the same time, experimentation with avalanche transceiver electronics and sensors was done. The ToF sensors for object avoidance and height approximation were tested in ice and snow conditions for the accuracy along with firmware requirements to allow for smooth integration with our final drone design. Simultaneously, the GPIO extender boards were tested with our avalanche transceiver to determine how to decode the signal direction and distance from it. This allowed us to verify the chips selected and begin the PCB design earlier on to account for long shipping and PCB fabrication lead times.

One of the major goals of the project schedule was to eliminate downtime by overlapping tasks whenever possible. This was again planned in the next phase once the initial Beta Flight testing was done. Once we had verified all of our sensor selections and while waiting for PCB fabrication, we focused on developing the waypoint logic, telemetry input handling, and autonomous flight behavior using ArduPilot. During this phase of the second half of the semester, we did the process of connecting each of the subsystems. This stage focused on testing integrating the custom PCB, telemetry sensors, ToF sensors, and avalanche beacon interface onto the drone platform. The autonomous flight software integration continued alongside integration testing of the electronics and created a resulting prototype.

The final weeks of the project focused on continuing system integration and indepth testing of the electronic systems before the final flight. This included simulating the waypoint calculations, and validating communication between the sensors, avalanche beacon, and flight controller. The full timeline and overlap between project tasks are shown in the Gantt chart included in Appendix C.

IX. Conclusion

The goal we had for SnowBat was to explore whether a lightweight autonomous drone could reduce avalanche victim search time and improve the effectiveness of companion rescue in the backcountry. Through the design and development of a custom drone platform integrating autonomous flight, avalanche beacon localization, terrain sensing, and custom electronics, the project successfully demonstrated the feasibility of many of the core technologies required for such a system.

Over the course of the project, substantial progress was made across the mechanical, electrical, and software subsystems. Mechanically, we developed a lightweight long enough range drone to be capable of supporting the full sensing and compute stack while remaining compact enough for backcountry use. Multiple frame and propeller configurations were evaluated, leading to the selection of a 7-inch long-range configuration that allowed for stability and efficiency for extended flight time. The design also evolved through multiple iterations of rapid prototyping, hardware testing, and repeated crash survivability during development.

On the electrical side, the project successfully integrated a custom PCB, GPIO expanders, Time-of-Flight sensors, GPS, flight controller, ESC, and avalanche beacon into a unified system. One of the largest technical accomplishments of the project was the successful decoding of a commercial avalanche beacon's LED display into machine-readable distance and directional information that could calculate an autonomous waypoint, all without modifying the beacon's internal firmware or RF circuitry. In parallel,

the ToF sensors were validated for operation on snow surfaces and demonstrated sufficient accuracy for terrain-following during the search.

From a software perspective, the project progressed from initial BetaFlight-based manual flight testing into a customized ArduPilot-based autonomous flight architecture. This allowed for autonomous waypoint setting through beacon decoding and coordinate transformation systems. Additionally, while fully autonomous end-to-end avalanche beacon tracking was not completed by the end of the project timeline, the majority of the supporting infrastructure and subsystem validation required for such a system was successfully demonstrated.

A major focus throughout the semester was reducing technical risk through basing our design off of iterative prototyping and doing as many things in parallel as possible. Sensor testing, PCB validation, flight PID tuning, waypoint software development, and avalanche beacon experimentation were intentionally overlapped to minimize downtime and maintain progress even during long lead times for shipping. This strategy allowed us to continuously validate and avoid major redesigns late in the project.

Although SnowBat remains an engineering prototype, the project demonstrates that a compact, deployable autonomous avalanche search drone is technically feasible using consumer hardware and custom software integration. Future work would focus on improving full autonomous search behavior, refining terrain-following and waypoint algorithms, and further reducing overall system size and weight. Additional testing in real avalanche terrain and extended cold-weather flight validation would also be needed if additional work was done.

Overall, SnowBat successfully achieved its primary goal of developing and validating the foundation of an autonomous avalanche search drone. The final prototype, software systems, custom electronics, testing results, and supporting documentation provide clear proof of the potential for autonomous drones to improve avalanche rescue response times in backcountry environments.

X. References

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XI. Appendix A: Relevant Standards

Since our proposed system incorporates an unmanned aerial vehicle weighing more than 250 grams, it must comply with applicable Federal Aviation Administration (FAA) regulations before it can be legally operated or sold for field use. At a minimum, the drone must be registered through FAA DroneZone, which currently requires a \$5 registration fee and remains valid for three years. Regardless of recreational or commercial use, each drone must be assigned a Remote ID through this process.

For commercial or professional SAR applications, such as use by tour guides, ski patrol teams, or resort personnel, the operator would typically need to hold an FAA Part 107 Remote Pilot Certificate. This requires passing a \$175 FAA knowledge exam at an approved testing center and demonstrates that the operator understands airspace rules, weather considerations, operating limitations, and safety procedures. In some cases, public agencies may opt to operate under a Public Aircraft Operation framework, which can allow certain government or emergency-response operations to bypass standard Part 107 requirements. However, this would depend on the specific organization, mission type, and legal classification of the operation, and would likely need to get manually approved if we or our operators choose to become incorporated as a public agency.

In addition to flight regulations, the system must account for intellectual property and commercial licensing requirements associated with integrating third-party avalanche rescue technology, i.e. our beacon. Since our design integrates BCA Tracker S avalanche beacons for a commercial SAR application, permission or licensing may be required from the manufacturer. We would need to directly contact Backcountry Access or register under its profession/commercial program [22] to determine whether we can incorporate the beacon legally into the final system.

XII. Appendix B: Bill of Materials

Part No.	Part Name	Manufacturer	Provider	Link	Unit Cost	Quantity	Total Cost
3317	VL53L0X TIME FLIGHT DIST STEMMA		Digikey	link	\$14.95	2	\$29.90
BC1712001010	BCA Tracker S Avalanche Beacon	BCA	SnowBig Deal	link	\$259.97	2	\$519.94
	BLITZ Mini F745 Stack	iFlight	iFlight	link	\$125.99	1	\$125.99
N/A	Professor Buccafusca Reimbursement	N/A	N/A	N/A	N/A	N/A	-\$259.97
	RP1 V2 ExpressLRS 2.4ghz Nano Receiver	Radiomaster	Amazon	link	\$20.99	4	\$20.99
	PYRODRONE SOURCE ONE 7" Long Range Frame - 6mm V0.2 Arms	Pyrodome	Pyrodron e	link	\$38.99	1	\$38.99
	Pyrodrome Hyperjuice 3000mAh 4S1P Li-ion Long Range Battery - XT60	Pyrodome	Pyrodron e	link	\$61.99	1	\$61.99
	16pcs HQProp Ethix S5 5x4x3 Smooth FPV Racing Props Poly Carbonate 5 Inch RC Drone Quad Tri-Blade Durable Propellers 5mm Shaft Hole	HQProp	Amazon	link	\$17.98	1	\$17.98
3317	VL53L0X TIME FLIGHT DIST STEMMA		Digikey	link	\$14.95	2	\$29.90
	T-Motor Velox V2306 V3.0 2550KV Brushless Drone Motor	Velox	rcdrone	link	\$25.00	2	\$50.00
	1680 Pcs M2 M3 M4 M5 Metric Hex Head Bolts and Nuts and Washers Set, Hex Socket Head Cap Screws Bolts and Nuts and Washers Assortment Kit, Black Machine Screws Set	KhjzgcTech	Amazon	link	\$13.98	1	\$13.98

	3 Pairs XT60 Plug Female and Male Connector,XT60 Connector Adapter with 5cm 12AWG	ZHOFONET	Amazon	link	\$7.99	1	\$7.99
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	Silicone Wire for RC Lipo Battery Cable Drone						
	LiPo Battery Charger - 2S 3S 4S RC Car Batteries Balance Chargers 18W 2A for LiPo(7.4-14.8V) LiHV(8.7-17.4V) with XH3P XH4P XH5P Plug for RC Battery Boat Airplane Truck Quadcopter Airsoft	Tosiicop Store	Amazon	link	\$19.99	1	\$19.99
	iFlight Blitz Mini M10 V2 GPS Module	iFlight	Pyrodron e	link	\$50.99	1	\$50.99
	Adafruit MCP23017 I2C GPIO Expander Breakout - STEMMA QT / Qwiic	Adafruit	Adafruit	link	\$5.95	3	\$17.85
	STEMMA QT / Qwiic JST SH 4-pin Cable - 100mm Long	Adafruit	Adafruit	link	\$0.95	10	\$9.50
SN74HC165N	IC SHIFT REGTR COMP 8BIT 16-PDIP	Texas Instruments	Digikey	link	\$1.42	5	\$7.10
3317	VL53L0X TIME FLIGHT DIST STEMMA		Digikey	link	\$14.95	2	\$29.90
	BrotherHobby Avenger 2808 Motor - 1350KV	BrotherHobby	getfpv	link	\$36.49	4	\$145.96
	APC 7x4 SF-B4 2-Blade Propellers (Set of 4)	APC	getfpv	link	\$7.49	2	\$14.98
	iFlight BLITZ Mini F745 Flight Controller - 20x20mm	iFlight	Pyrodron e	link	\$76.99	1	\$76.99
	RadioMaster RP1 V2 ELRS 2.4GHz Nano Receiver w/ UFL Antenna	Radiomaster	Pyrodron e	link	\$24.99	1	\$24.99
	EVE 33V 18650 3200mAh 10A Battery	EVE	18650Bat teryStore	link	\$2.54	4	\$10.16
	APC 7x4 SF-B4 2-Blade Propellers (Set of 4)	APC	getfpv	link	\$7.49	2	\$14.98
	GPIO Ribbon Cable 2x10 IDC Cable - 20 pins 12" long	Adafruit	Adafruit	link	\$1.95	5	\$9.75

3020-20-030 0-00-TR	CONN HEADER SMD 20POS 2.54MM	CNC Tech	Digikey	link	\$1.64	5	\$8.20
800AWSP9 M2QE	SWITCH PUSH SPST-NO 1A 120V	E-Switch	Digikey	link	\$3.95	5	\$19.75
	CHANZON 10 pcs High Power Led Chip 1W Warm White 3000K - 3500K (Input 300mA - 350mA / DC 3V - 3.4V / 1 Watt) Super Bright Intensity SMD COB Light Emitter Components Diode 1 W Bulb Bead DIY Lighting	Chanzon	Amazon	link	\$6.99	1	\$6.99
	M3 locknuts	Einundey	Amazon	link	\$5.29	1	\$5.29
	Cable ties	HAVE ME TD Store	Amazon	link	\$2.99	1	\$2.99
	1.75mm TPU filament	Sunlu	Amazon	link	\$23.49	1	\$23.49
	More M3 Screw	TOOINKCV	Amazon	link	\$6.88	1	\$6.88
	PCB	JLCPCB	JLCPCB	N/A	\$132.89	1	\$132.89

XIII. Appendix C: Gantt Chart

